



FORT STREET HIGH SCHOOL

Student Number: \_\_\_\_\_

Teacher: \_\_\_\_\_

Class: \_\_\_\_\_

# 2018

## HIGHER SCHOOL CERTIFICATE COURSE

### ASSESSMENT TASK 3: TRIAL HSC

## Mathematics Extension 1

**Time allowed: 2 hours**

(plus 5 minutes reading time)

| Syllabus Outcomes | Assessment Area Description and Marking Guidelines                                                                                                        | Questions |
|-------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|
|                   | Chooses and applies appropriate mathematical techniques in order to solve problems effectively                                                            | 1-10      |
| HE2, HE4          | Manipulates algebraic expressions to solve problems from topic areas such as inverse functions, trigonometry, polynomials, permutations and combinations. | 11, 12    |
| HE3, HE5<br>HE6   | Uses a variety of methods from calculus to investigate mathematical models of real life situations, such as projectiles, kinematics and growth and decay  | 13        |
| HE7               | Synthesises mathematical solutions to harder problems and communicates them in appropriate form                                                           | 14        |

### Total Marks 70

#### Section I 10 marks

Multiple Choice, attempt all questions,  
Allow about 15 minutes for this section

#### Section II 60 Marks

Attempt Questions 11-14,  
Allow about 1 hour 45 minutes for this section

### General Instructions:

- Questions 11-14 are to be started in a new booklet.
- The marks allocated for each question are indicated.
- In Questions 11 – 14, show relevant mathematical reasoning and/or calculations.
- Marks may be deducted for careless or badly arranged work.
- Board – approved calculators may be used.

| Section I  | Total 10 | Marks |
|------------|----------|-------|
| Q1-Q10     |          |       |
| Section II | Total 60 | Marks |
| Q11        | /15      |       |
| Q12        | /15      |       |
| Q13        | /15      |       |
| Q14        | /15      |       |
|            | Percent  |       |

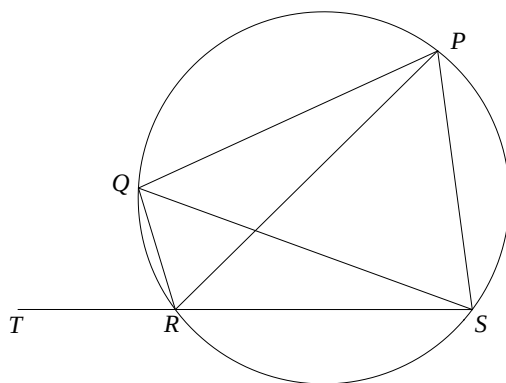
**SECTION I (One mark each)**

**Answer questions 1 to 10 on the multiple choice answer sheet.**

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- 1 The number plates for motor vehicles consist of 2 letters, 2 numbers and then 2 letters such as AB01CD. How many different number plates are possible?
- (A) 135 200  
(B) 270 400  
(C) 37 015 056  
(D) 45 697 600
- 2 What is the exact value of  $\tan 75^\circ$ ?
- (A)  $2 - \sqrt{3}$   
(B)  $4 - \sqrt{3}$   
(C)  $2 + \sqrt{3}$   
(D)  $4 + \sqrt{3}$
- 3 A curve has parametric equations  $x = \frac{2}{t}$  and  $y = 2t^2$ .  
What is Cartesian equation of this curve?
- (A)  $y = \frac{4}{x}$   
(B)  $y = \frac{8}{x}$   
(C)  $y = \frac{4}{x^2}$   
(D)  $y = \frac{8}{x^2}$
- 4 What are the coordinates of the point P that divides internally the interval joining the points A(1, 2) and B(7, 5) in the ratio 2:1?
- (A) (3, 3)  
(B) (3, 4)  
(C) (5, 3)  
(D) (5, 4)

- 5 What is the solution to the inequality  $\frac{3}{x-2} \leq 4$ ?
- (A)  $x < -2$  and  $x \geq -\frac{11}{4}$
- (B)  $x > -2$  and  $x \leq -\frac{11}{4}$
- (C)  $x < 2$  and  $x \geq \frac{11}{4}$
- (D)  $x > 2$  and  $x \leq \frac{11}{4}$
- 6 What is the acute angle to the nearest degree between the lines  $y = 1 - 3x$  and  $4x - 6y - 5 = 0$ ?
- (A)  $15^\circ$
- (B)  $38^\circ$
- (C)  $52^\circ$
- (D)  $75^\circ$
- 7 PQRS is a cyclic quadrilateral. SR is produced to T and  $\angle PRS = \angle QRT$ .



Why is  $\angle PQS = \angle PRS$ ?

- (A) Angle at the circumference is equal to the angle in the alternate segment.
- (B) Angle between the tangent and a chord is equal to the angle in the alternate segment.
- (C) Angle between the two chords in the same segment are equal.
- (D) Angles in the same segment standing on the same arc are equal.
- 8 Consider the polynomial  $P(x) = 3x^3 + 3x + a$ .  
If  $x - 2$  is a factor of  $P(x)$ , what is the value of  $a$ ?
- (A)  $-30$
- (B)  $-18$
- (C)  $18$
- (D)  $30$

**9** Which of the following is an expression for  $\int \cos^2 2x \, dx$ ?

(A)  $x - \frac{1}{4} \sin 4x + c$

(B)  $x + \frac{1}{4} \sin 4x + c$

(C)  $\frac{x}{2} - \frac{1}{8} \sin 4x + c$

(D)  $\frac{x}{2} + \frac{1}{8} \sin 4x + c$

**10** A particle moving in a straight line obeys  $v^2 = -x^2 + 2x + 8$  where  $x$  is its displacement from the origin in metres and  $v$  is its velocity in m/s. The motion is simple harmonic. What is the amplitude?

(A)  $2\pi$  metres

(B) 3 metres

(C) 8 metres

(D) 9 metres

**SECTION II (15 marks for each question.)**

**Answer each question in the appropriate booklet. Extra writing booklets are available.**

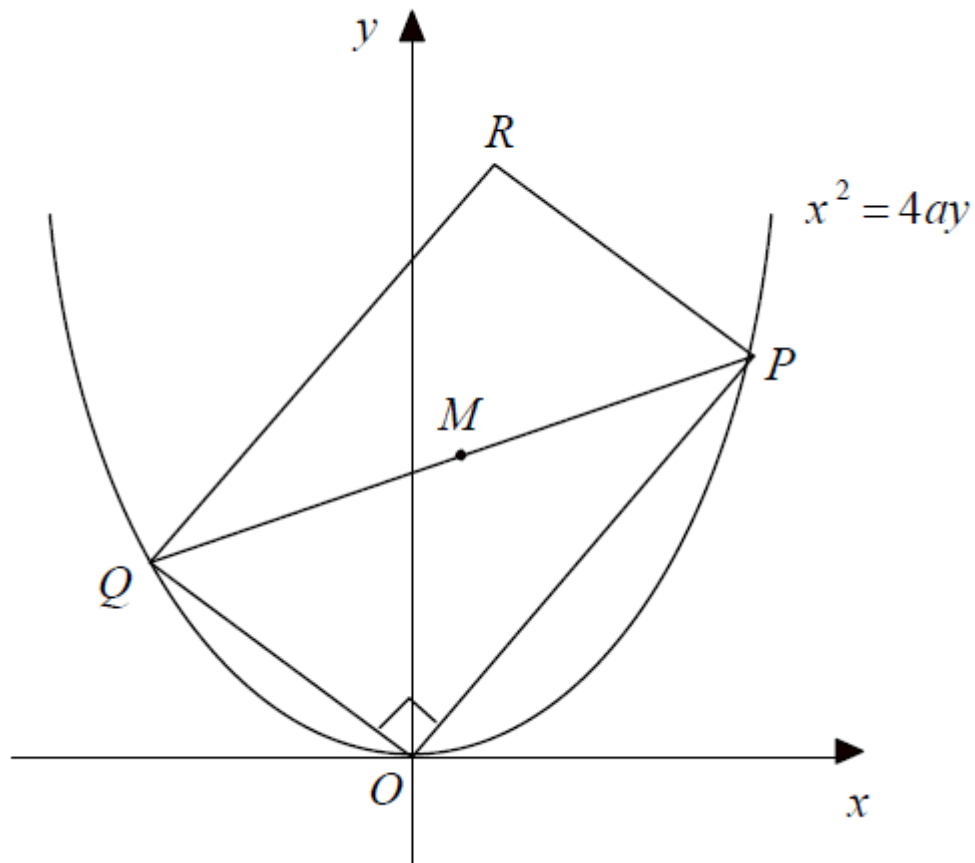
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**Question 11:**     *Use a separate writing booklet .*

- a) The letters of the word MOUSE are to be rearranged.
- i.    How many arrangements are there which start with the letter M and end with the letter E? 1
  - ii.   How many arrangements are there in which the vowels are grouped together? (A vowel is one of the letters A, E, I, O, U) 1
  - iii.   How would your answers to parts (i) and (ii) change if the given word had been MOOSE instead of MOUSE? 1
- b) Write  $7.\dot{1}\dot{2}$  as the sum of an infinite series. Hence write  $7.\dot{1}\dot{2}$  as a mixed fraction. 2
- c) Given the function  $g(x) = \sqrt{x+2}$  and that  $g^{-1}(x)$  is the inverse function of  $g(x)$ , find  $g^{-1}(5)$ . 2
- d) Solve the equation  $1 + \cos 2x = \sin 2x$  for  $0 \leq x \leq 2\pi$ . 4

Question 11 continues over the page.

e)



$P(2ap, ap^2)$  and  $Q(2aq, aq^2)$  are two points which move on the parabola  $x^2 = 4ay$  such that  $\angle POQ = 90^\circ$ , where  $O(0,0)$  is the origin.  $M\left(a(p+q), \frac{a}{2}(p^2+q^2)\right)$  is the midpoint of  $PQ$ .  $R$  is the point such that  $OPRQ$  is a rectangle.

- i. Show that  $pq = -4$ . 1
- ii. Show that  $R$  has coordinates of  $(2a(p+q), a(p^2+q^2))$ . 1
- iii. Find the equation of the locus of  $R$ . 2

**Question 12:**      *Use a separate writing booklet.*

- a) The polynomial equation  $P(x) = 3x^3 - 2x^2 + x - 4$  has roots  $\alpha, \beta$  and  $\gamma$ . Find the exact value of  $\frac{1}{\alpha\beta} + \frac{1}{\beta\gamma} + \frac{1}{\alpha\gamma}$ . 2
- b) Find  $\lim_{h \rightarrow 0} \left( \frac{\cos 2h - 1}{h} \right)$ . 2
- c) .
- i. Sketch the graph of  $y = |2 - x|$ . 1
- ii. Using this graph, or otherwise, find the solution to  $|2 - x| < x$ . 2
- d) Use the method of Mathematical induction to show that  $2 + 5 + 8 + \dots + (3n - 1) = \frac{n(3n + 1)}{2}$  3
- e) A plate is initially heated to  $55^\circ\text{C}$ , and it then cools to  $41^\circ\text{C}$  in 10 minutes. Assume Newtons Law of Cooling  $\frac{dT}{dt} = -k(T - S)$  applies, where  $S$  is the surrounding temperature. If the surrounding temperature is  $22^\circ\text{C}$  :
- i. Show that the temperature  $T$  is given by  $T = 22 + 33e^{\frac{-t}{10} \ln\left(\frac{33}{19}\right)}$ . 2
- ii. Find the temperature after 25 minutes (to the nearest degree). 1
- iii. Find the time for the plate to cool to  $25^\circ\text{C}$  (to 1 decimal place). 2

**Question 13:** Use a separate writing booklet.

- a) Find  $\frac{d}{dx}(e^{-x} \cos^{-1} x)$ . 2
- b) By making the substitution  $u = x - 2$ , evaluate  $\int_4^5 \frac{x(x-4)}{(x-2)} dx$ . 3
- c) For the polynomial  $P(x) = x^3 + 2x^2 - 15x - 36$ ,
- i. Factorize  $P(x)$  fully over the real numbers. 3
- ii. Hence solve  $x^3 + 2x^2 - 15x - 36 < 0$ . 2
- d) Wheat falls from an auger onto a conical pile at the rate of  $20 \text{ cm}^3/\text{s}$ . The radius of the base of the pile is always equals to half its height.
- i. Show that  $V = \frac{\pi}{12} h^3$  and hence find  $\frac{dh}{dt}$ . 2
- ii. Find the rate, in terms of  $\pi$ , at which the pile is rising when it is 8cm high. 1
- iii. Find the time, to 1 decimal place, for the pile to reach a height of 8cm. 2

**Question 14:** Use a separate writing booklet.

- a) A particle moves in a straight line on the  $x$  axis. At time  $t$  its velocity is  $v$  and its acceleration is  $\ddot{x}$ . Given  $\ddot{x} = 4 - 4x$  and initially  $x = 3$  when  $v^2 = 20$ :
- i. Show that the motion is SHM and state the centre of motion and the value of  $n$ . 2
  - ii. Hence show that  $v^2 = 32 + 8x - 4x^2$ . 1
  - iii. Use this expression for  $v^2$  to find the possible values for  $x$ . 1
  - iv. Describe the motion of the particle until it first stops if initially  $v = -2\sqrt{5}$ . 1
- b)
- i. Differentiate  $x \sin^{-1} x$ , and hence find an equation to evaluate  $\int \sin^{-1} x \, dx$ . 2
  - ii. Hence evaluate  $\int_0^1 \sin^{-1} x \, dx$ . 2
- c) An object is projected from the origin  $O$  with initial speed  $U$  m/s at an angle of elevation of  $\alpha$ . At the same instant another object is projected from a point  $A$  which is  $h$  units above the origin  $O$ . The second object is projected with initial speed  $V$  m/s at an angle of elevation of  $\beta$ , where  $\beta < \alpha$ . Both objects move freely under gravity in the same plane.
- i. Given that the equations of motion for the object projected from the origin are:
$$x = Ut \cos \alpha \quad y = Ut \sin \alpha - \frac{1}{2}gt^2,$$
write down the equations of motion for the object projected from the point  $A$ . 2
  - ii. If the objects collide  $T$  seconds after they are simultaneously projected, show that  $T = \frac{h \cos \beta}{U \sin(\alpha - \beta)}$  4

**End of examination.**

|   |                                                                                                                                                                                                                                                                                                                  |           |
|---|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|
| 1 | $\text{Number of arrangements} = 26^2 \times 10^2 \times 26^2$ $= 45\,697\,600$                                                                                                                                                                                                                                  | 1 Mark: D |
| 2 | $\tan 75^\circ = \tan(45^\circ + 30^\circ)$ $= \frac{\tan 45^\circ + \tan 30^\circ}{1 - \tan 45^\circ \tan 30^\circ}$ $= \frac{1 + \frac{1}{\sqrt{3}}}{1 - \frac{1}{\sqrt{3}}} \times \frac{\sqrt{3}}{\sqrt{3}}$ $= \frac{\sqrt{3} + 1}{\sqrt{3} - 1} \times \frac{\sqrt{3} + 1}{\sqrt{3} + 1}$ $= 2 + \sqrt{3}$ | 1 Mark: C |
| 3 | $x = \frac{2}{t} \text{ or } t = \frac{2}{x}$ <p>Substitute <math>\frac{2}{x}</math> for <math>x</math> into <math>y = 2t^2</math></p> $y = 2\left(\frac{2}{x}\right)^2$ $= \frac{8}{x^2}$                                                                                                                       | 1 Mark: D |
| 4 | $x = \frac{mx_2 + nx_1}{m + n} \quad y = \frac{my_2 + ny_1}{m + n}$ $= \frac{2 \times 7 + 1 \times 1}{2 + 1} = 5 \quad = \frac{2 \times 5 + 1 \times 2}{2 + 1} = 4$ <p>The coordinates of P are (5, 4)</p>                                                                                                       | 1 Mark: D |
| 5 | $(x - 2)^2 \times \frac{3}{(x - 2)} \leq 4 \times (x - 2)^2$ $(x - 2)3 \leq 4(x - 2)^2 \quad x \neq 2$ $(x - 2)(3 - 4x + 8) \leq 0$ $(x - 2)(11 - 4x) \leq 0$ $x < 2 \text{ and } x \geq \frac{11}{4}$                                                                                                           | 1 Mark: C |

|    |                                                                                                                                                                                                                                                                                                                                                                                    |           |
|----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|
| 6  | <p>For <math>y = 1 - 3x</math> then <math>m_1 = -3</math></p> <p>For <math>4x - 6y - 5 = 0</math> then <math>m_2 = \frac{2}{3}</math></p> $\tan \theta = \left  \frac{m_1 - m_2}{1 + m_1 m_2} \right $ $= \left  \frac{-3 - \frac{2}{3}}{1 + -3 \times \frac{2}{3}} \right $ $= \left  \frac{-\frac{11}{3}}{-1} \right $ $= \frac{11}{3}$ $\theta = 74.7448813.. \approx 75^\circ$ | 1 Mark: D |
| 7  | <p><math>\angle PQS = \angle PRS</math></p> <p>(angles in the same segment standing on the same arc are equal).</p>                                                                                                                                                                                                                                                                | 1 Mark: D |
| 8  | <p><math>P(x) = 3x^3 + 3x + a</math></p> <p><math>P(2) = 3(2)^3 + 3(2) + a = 0</math> (<math>x - 2</math> is a factor of <math>P(x)</math>)</p> <p><math>24 + 6 + a = 0</math></p> <p><math>a = -30</math></p>                                                                                                                                                                     | 1 Mark: A |
| 9  | $\int \cos^2 2x dx = \int \frac{1}{2}(1 + \cos 4x) dx$ $= \frac{1}{2} \left[ x + \frac{1}{4} \sin 4x \right] + c$ $= \frac{x}{2} + \frac{1}{8} \sin 4x + c$                                                                                                                                                                                                                        | 1 Mark: D |
| 10 | <p><math>v^2 = -x^2 + 2x + 8</math></p> <p><math>= 1^2(8 + 2x - x^2)</math></p> <p><math>= 1^2(9 - (x - 1)^2)</math></p> <p><math>= n^2(a^2 - x^2)</math></p> <p><math>a^2 = 9</math></p> <p><math>a = 3</math></p> <p>Amplitude is 3 metres</p>                                                                                                                                   | 1 Mark: B |

**Question 11:**

- a)
- $1 \times 3! \times 1$  ❶  
= 6
  - Grouping vowels is one item, leaves  $3!$  ways to the vowel group and the 2 consonants. But there are  $3!$  arrangements of the vowels, hence  $3! \times 3! = 36$  ways. ❶
  - Both answers would have OU and UO count as one, hence both answers would be divided by 2. ❶

- b)  $7.\dot{1}\dot{2} = 7 + 0.12 + 0.0012 + 0.000012 + \dots$  which is a GP with  $a = 0.12, r = 0.01$  ❶. Thus

$$\begin{aligned} 7.\dot{1}\dot{2} &= 7 + \frac{a}{1-r} \\ &= 7 + \frac{0.12}{1-0.01} \\ &= 7\frac{4}{33} \text{ ❶} \end{aligned}$$

- c)  $g^{-1}(x)$ :  
 $x = \sqrt{y+2}$   
 $x^2 = y+2$   
 $y = x^2 - 2$   
 $g^{-1}(x) = x^2 - 2$  ❶  
 $g^{-1}(5) = 5^2 - 2$   
 $= 23$  ❶

- d)  $1 + \cos 2x = \sin 2x$   
 $2 \cos^2 x = 2 \sin x \cos x$  ❶  
 $\cos^2 x - \sin x \cos x = 0$   
 $\cos x (\cos x - \sin x) = 0$  ❶

Hence:

$$\begin{aligned} \cos x &= 0 & \cos x - \sin x &= 0 \\ x &= \frac{\pi}{2}, \frac{3\pi}{2} \text{ ❶} & \cos x &= \sin x \\ & & 1 &= \tan x \\ & & x &= \frac{\pi}{4}, \frac{5\pi}{4} \text{ ❶} \end{aligned}$$

$$\therefore x = \frac{\pi}{4}, \frac{\pi}{2}, \frac{5\pi}{4}, \frac{3\pi}{2} \quad (0 \leq x \leq 2\pi)$$

- e)
- Gradient OP:  $m_{OP} = \frac{ap^2 - 0}{2ap - 0}$  and gradient OQ:  $m_{OQ} = \frac{aq^2 - 0}{2aq - 0}$   
 $= \frac{p}{2} \qquad \qquad \qquad = \frac{q}{2}$

Marking

Comments

Reasoning must be clear

Must show a GP as reqd by question or no marks

Question asked how the answer changed, not what the new answer was.

Many students did not follow the instructions and hence gained no marks.

Dividing by  $2 \cos x$  was a common error – this lost one set of solutions!

As many students attempted the more complex  $R \sin(2x - \theta)$  or  $R \cos(2x + \theta)$  transformations, they are duplicated at the end of the solutions.

Many found  $p$  and  $q$  for gradients (the tangents) in error.

As  $OP \perp OQ$ ,  $m_{OP} \times m_{OQ} = -1$

$$\frac{p}{2} \times \frac{q}{2} = -1 \text{ ①}$$

$$pq = -4$$

- ii. The diagonals of the rectangle bisect each other, hence  $M$  is also the midpoint of  $OR$ . Thus

$$x_M = \frac{x_R + 0}{2} \quad y_M = \frac{y_R + 0}{2}$$

$$x_R = 2x_M \quad y_R = 2y_M \text{ ①}$$

$$= 2a(p+q) \quad = 2 \cdot \frac{a}{2}(p^2 + q^2)$$

$$= a(p^2 + q^2)$$

So  $R$  is  $(2a(p+q), a(p^2 + q^2))$

- iii.  $x_R = 2a(p+q)$  and  $y_R = a(p^2 + q^2)$

Hence  $(p+q) = \frac{x_R}{2a}$ , and then

$$y_R = a(p^2 + q^2 + 2pq - 2pq)$$

$$= a((p+q)^2 - 2pq)$$

$$= a\left(\left(\frac{x_R}{2a}\right)^2 - 2(-4)\right) \text{ ①}$$

$$= a\left(\frac{x_R^2}{4a^2} + 8\right)$$

$$= \frac{x_R^2}{4a} + 8a$$

$$\text{So } x^2 = 4a(y - 8a) \text{ ①}$$

Marking

Comments

Correct use of the midpoint formula.

Both substitutions correct.

Many lengthy and convoluted methods used for this one mark – usually without success.

Many expanded the second line to  $a(p+q)^2 - 2pq$ , instead of  $a(p+q)^2 - 2apq$  (failing to multiply the last term by  $a$ ), thus losing a mark.

**Question 12:**

$$\begin{aligned}
 \text{a)} \quad \alpha + \beta + \gamma &= \frac{2}{3}; \quad \alpha\beta\gamma = \frac{4}{3} \\
 \frac{1}{\alpha\beta} + \frac{1}{\beta\gamma} + \frac{1}{\alpha\gamma} &= \frac{\alpha + \beta + \gamma}{\alpha\beta\gamma} \text{ ①} \\
 &= \frac{\frac{2}{3}}{\frac{4}{3}} \\
 &= \frac{1}{2} \text{ ①}
 \end{aligned}$$

$$\begin{aligned}
 \text{b)} \quad \lim_{h \rightarrow 0} \left( \frac{\cos 2h - 1}{h} \right) &\times \left( \frac{\cos 2h + 1}{\cos 2h + 1} \right) \\
 &= \lim_{h \rightarrow 0} \left( \frac{\cos^2 2h - 1}{h(\cos 2h + 1)} \right) \\
 &= \lim_{h \rightarrow 0} \left( \frac{-\sin^2 2h}{h(\cos 2h + 1)} \right) \\
 &= \lim_{h \rightarrow 0} \left( \frac{\sin 2h}{2h} \times \frac{-2 \sin 2h}{(\cos 2h + 1)} \right) \text{ ①} \\
 &= \lim_{h \rightarrow 0} \left( \frac{\sin 2h}{2h} \right) \times \lim_{h \rightarrow 0} \left( \frac{-2 \sin 2h}{\cos 2h + 1} \right) \\
 &= 1 \times 0 \\
 &= 0 \text{ ①}
 \end{aligned}$$

Or

$$\begin{aligned}
 \lim_{h \rightarrow 0} \left( \frac{\cos 2h - 1}{h} \right) &= \lim_{h \rightarrow 0} \left( \frac{1 - 2 \sin^2 h - 1}{h} \right) \text{ ①} \\
 &= \lim_{h \rightarrow 0} \left( \frac{-2 \sin^2 h}{h} \right) \\
 &= 2 \lim_{h \rightarrow 0} \left( \frac{\sin h}{h} \right) \cdot \lim_{h \rightarrow 0} (\sin h) \\
 &= 2.1.0 \\
 &= 0 \\
 &\text{ ①}
 \end{aligned}$$

Or

$$\begin{aligned}
 &2 \lim_{h \rightarrow 0} \left( \frac{\sin h}{h} \right) \cdot \lim_{h \rightarrow 0} (\sin h) \\
 &= 2 \lim_{h \rightarrow 0} \left( \frac{\sin h}{h} \right) \cdot \lim_{h \rightarrow 0} \left( \frac{\sin h}{h} \right) \lim_{h \rightarrow 0} h \\
 &= 2.1.1.0 = 0 \\
 &\neq h
 \end{aligned}$$

Marking

Alternative  
methods possible  
here

Comments

Keep it simple. If you can't memorise

$$\frac{1}{\alpha\beta} + \frac{1}{\beta\gamma} + \frac{1}{\alpha\gamma} = \frac{\alpha + \beta + \gamma}{\alpha\beta\gamma}$$

..then find the common denominator.

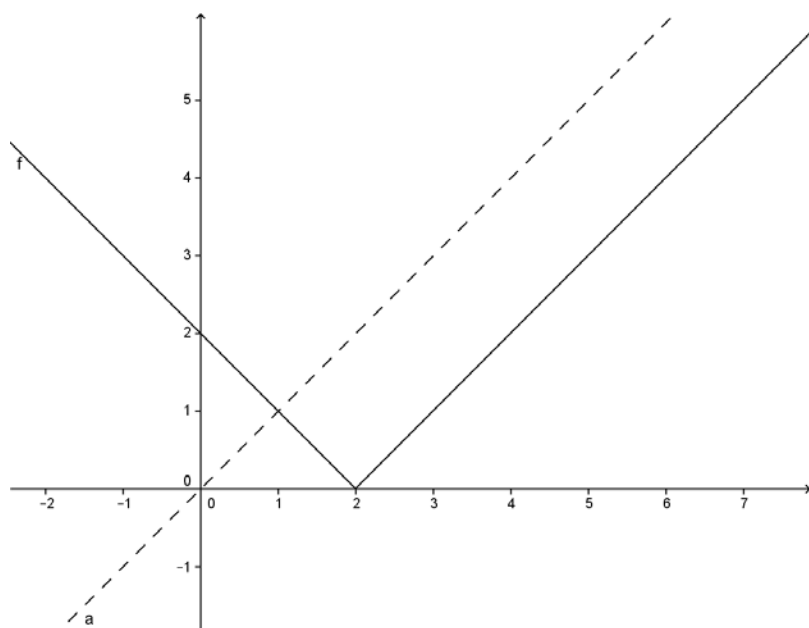
$$\frac{\sin^2 h}{h} \neq \frac{\sin h}{h} \frac{\sin h}{h}$$

Remember

$$\lim_{h \rightarrow 0} h = 0 \neq h$$

c)

i.



● sketch of  $y = |2 - x|$  correct.

- ii. Comparing to  $y = x$  (dashed line above), the graphs cross at (1,1) ●, hence  $|2 - x| < x$  when  $x > 1$ . ●

Marking

Comments

Label the axes  $x$  &  $y$ .

Use arrows to indicate the positive direction and arrows on your rays/lines to indicate that they're not line segments.

Label intercepts with just the intercept value, not the full coordinates of the intercepts.

This should be explicit.

If solving algebraically, you must test your solutions. Solving algebraically often generates solutions which aren't valid, so they must be tested and eliminated.

d) Step 1: show true for  $n = 1$

$$\begin{aligned} LHS &= 2 \quad RHS = \frac{n(3n+1)}{2} \\ &= \frac{1(3+1)}{2} \\ &= 2 \\ &= LHS \end{aligned}$$

Hence true for  $n = 1$ .

Step 2: Assume true for  $n = k$ :

$$\text{i.e. assume } 2 + 5 + 8 + \dots + (3k - 1) = \frac{k(3k + 1)}{2}$$

Then show true for  $n = k + 1$

i.e. show

$$2 + 5 + 8 + \dots + (3k - 1) + (3(k + 1) - 1) = \frac{(k + 1)(3(k + 1) + 1)}{2}$$

❶ for setting up correctly.

Step 3:

$$\begin{aligned} LHS &= 2 + 5 + 8 + \dots + (3k - 1) + (3(k + 1) - 1) \\ &= \frac{k(3k + 1)}{2} + (3(k + 1) - 1) \\ &\quad \text{(using the assumption ❶)} \\ &= \frac{k(3k + 1)}{2} + 3k + 2 \\ &= \frac{k(3k + 1) + 6k + 4}{2} \\ &= \frac{3k^2 + k + 6k + 4}{2} \\ &= \frac{3k^2 + 7k + 4}{2} \\ &= \frac{3k^2 + 3k + 4k + 4}{2} \\ &= \frac{3k(3k + 1) + 4(k + 1)}{2} \\ &= \frac{(k + 1)(3k + 4)}{2} \\ &= \frac{(k + 1)(3(k + 1) + 1)}{2} \quad \text{as reqd. ❶ (correct algebra to result)} \end{aligned}$$

Hence, by Mathematical Induction,

$$2 + 5 + 8 + \dots + (3n - 1) = \frac{n(3n + 1)}{2}$$

Marking

Comments

Do not expand and simplify at this stage.

This should be explicit.

Most people stopped here. One line early.

Show the RHS result here.

e)

$$\text{i.} \quad \frac{dT}{dt} = -k(T - S)$$

$$\frac{dT}{(T - S)} = -k dt$$

Integrating:

$$\ln(T - S) = -kt + c$$

$$T - S = e^{-kt+c}$$

$$= Ae^{-kt}$$

$$T = S + Ae^{-kt}$$

$$T = 22 + Ae^{-kt} \quad (\text{as } S = 22).$$

When  $t = 0, T = 55$ , hence  $55 = 22 + A.1$ 

$$A = 33 \text{ ①}$$

When  $t = 10, T = 41$ , hence

$$41 = 22 + 33e^{-10k}$$

$$\frac{19}{33} = e^{-10k}$$

$$-10k = \ln\left(\frac{19}{33}\right)$$

$$k = \frac{-1}{10} \ln\left(\frac{19}{33}\right)$$

$$= \frac{1}{10} \ln\left(\frac{33}{19}\right) \text{ ①}$$

Thus  $T = 22 + 33e^{\frac{-t}{10} \ln\left(\frac{33}{19}\right)}$  as reqd.ii.  $t = 25$ :

$$T = 22 + 33e^{\frac{-25}{10} \ln\left(\frac{33}{19}\right)}$$

$$= 30.3^\circ \text{C} \text{ ①}$$

Marking

Comments

Do not try to show that the solution fits the model by differentiating it's RHS.

Note:  $S$  is given – students just need to interpret this correctly.

Many students stopped at

$$k = \frac{-1}{10} \ln\left(\frac{19}{33}\right)$$

and fudged.

Make  $k$  look positive (distribute the negative into the log) if you are working with a model which looks like this:

$$P = B + Ae^{-kt}$$

iii.  $T = 25$ :

$$25 = 22 + 33e^{\frac{-t}{10}\ln\left(\frac{33}{19}\right)}$$

$$\frac{3}{33} = e^{\frac{-t}{10}\ln\left(\frac{33}{19}\right)}$$

$$\ln\left(\frac{1}{11}\right) = \frac{-t}{10}\ln\left(\frac{33}{19}\right) \text{ ①}$$

$$\frac{10\ln\left(\frac{1}{11}\right)}{\ln\left(\frac{33}{19}\right)} = -t$$

$$t = \frac{10\ln(11)}{\ln\left(\frac{33}{19}\right)}$$

$$= 43.43473528$$

$$\approx 43.4 \text{ min } \text{ ①}$$

Marking

Comments

**Question 13:**

$$\begin{aligned} \text{a) } \frac{d}{dx}(e^{-x} \cos^{-1} x) \\ = -e^{-x} \cos^{-1} x - \frac{e^{-x}}{\sqrt{1-x^2}} \end{aligned}$$

①                      ①

$$\begin{aligned} \text{b) With } u = x - 2 \text{ and } x = 5 \quad x = 4 \\ du = dx \quad u = 3 \quad u = 2 \end{aligned}$$

Then

$$\begin{aligned} I &= \int_4^5 \frac{x(x-4)}{(x-2)} dx \\ &= \int_2^3 \frac{(u+2)(u-2)}{u} du \\ &= \int_2^3 \frac{u^2 - 4}{u} du \\ &= \int_2^3 u - \frac{4}{u} du \quad \text{①} \\ &= \left[ \frac{u^2}{2} - 4 \ln(u) \right]_2^3 \quad \text{①} \\ &= \frac{9}{2} - 4 \ln 3 - \left( \frac{4}{2} - 4 \ln 2 \right) \\ &= \frac{5}{2} - 4 \ln \left( \frac{3}{2} \right) \quad \text{①} \end{aligned}$$

Marking

Comments

Well done.

**Note:**

$$\cos^{-1} x \neq \frac{1}{\cos x}$$

Generally well done.

$\frac{5}{2} + 4 \ln \frac{2}{3}$  also correct.

Evaluate means 'give exact answer'

i.e. don't approximate.

c)  $P(x) = x^3 + 2x^2 - 15x - 36$

i. Testing:

$$P(1) = -48 \quad P(-1) = -20 \quad P(2) = -50 \quad P(-2) = 6$$

$$P(3) = -36 \quad P(-3) = 0$$

Hence  $x+3$  is a factor of  $P(x)$ . ①

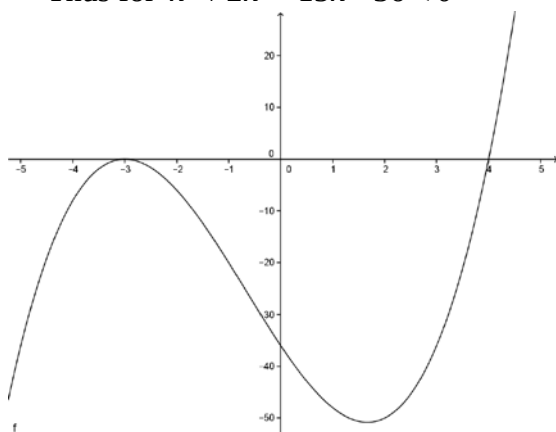
$$\begin{array}{r} x^2 - 1x - 12 \\ x+3 \overline{) x^3 + 2x^2 - 15x - 36} \quad \text{①} \\ \underline{x^3 + 3x^2} \phantom{- 15x - 36} \\ -x^2 - 15x \phantom{- 36} \\ \underline{-x^2 - 3x} \phantom{- 36} \\ \phantom{-x^2 - } -12x - 36 \\ \underline{-12x - 36} \\ \phantom{-12x - } 0 \end{array}$$

Now  $x^2 - 1x - 12$

$$= (x-4)(x+3)$$

Hence  $P(x) = (x-4)(x+3)^2$  ①

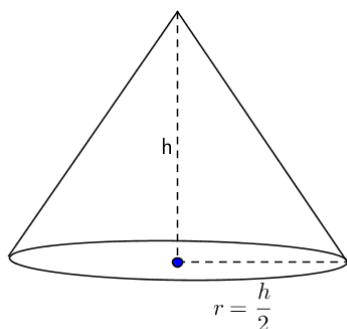
ii. Thus for  $x^3 + 2x^2 - 15x - 36 < 0$



① (graph or equivalent justification)

Hence  $x < -3$  and  $-3 < x < 4$  ①

d) Diagram:



Marking

Variants possible depending on root chosen.

Comments

Should show this working but was not penalised if it wasn't. Generally very well done.

A graph or testing regions needed to be shown to gain marks.

$x < -3$ ,  $x \neq 4$  also acceptable answer.

## Marking

## Comments

$$\begin{aligned} \text{i. } V &= \frac{1}{3}\pi r^2 h \\ &= \frac{1}{3}\pi \left(\frac{h}{2}\right)^2 h \text{ ①} \\ &= \frac{\pi}{12} h^3 \end{aligned}$$

$$\therefore \frac{dV}{dh} = \frac{\pi}{4} h^2$$

Then

$$\frac{dV}{dt} = \frac{dV}{dh} \times \frac{dh}{dt}$$

$$20 = \frac{\pi}{4} h^2 \times \frac{dh}{dt}$$

$$\frac{dh}{dt} = \frac{80}{\pi h^2} \text{ ①}$$

$$\begin{aligned} \text{ii. } \frac{dh}{dt} &= \frac{80}{\pi h^2} \\ &= \frac{80}{\pi \cdot 8^2} \\ &= \frac{5}{4\pi} \text{ ①} \end{aligned}$$

$$\text{iii. } \frac{dh}{dt} = \frac{80}{\pi h^2}$$

$$\pi h^2 dh = 80 dt$$

Integrating:

$$\frac{\pi}{3} h^3 = 80t + c$$

$$t = 0, h = 0, \text{ hence } c = 0.$$

$$\therefore \frac{\pi}{3} h^3 = 80t$$

$$t = \frac{\pi h^3}{240} \text{ ①}$$

When  $h = 8$ :

$$\begin{aligned} t &= \frac{\pi \cdot 8^3}{240} \\ &= \frac{32\pi}{15} \\ &= 6.7 \text{ sec. } \text{ ①} \end{aligned}$$

Needed to show this line as it was a 'show' question.

Well done.

Well done.

Other valid methods accepted.

One should show the evaluation of the integration constant,  $c$  but not penalised.

A common error was to start with  $\frac{dh}{dt} = \frac{5}{4\pi}$  which is a specific case rather than the general case. This led to getting an incorrect answer of 20.1 secs.

**Question 14:**a) From  $\ddot{x} = 4 - 4x$ 

$$\begin{aligned} \text{i. } \ddot{x} &= -4(x-1) \\ &= -2^2(x-1) \end{aligned}$$

which is of the form  $\ddot{x} = -n^2x$  ❶, hence the motion is SHM with  $n = 2$  and centre of motion at  $x = 1$ . ❶

$$\begin{aligned} \text{ii. } \frac{d\left(\frac{1}{2}v^2\right)}{dx} &= 4 - 4x \\ \frac{1}{2}v^2 &= 4x - 2x^2 + c \\ v^2 &= 8x - 4x^2 + c \end{aligned}$$

When  $x = 3$ ,  $v^2 = 20$ :

$$20 = 8 \cdot 3 - 4 \cdot 3^2 + c$$

$$20 = 24 - 36 + c$$

$$c = 32 \quad \text{❶ hence}$$

$$v^2 = 32 + 8x - 4x^2, \text{ as reqd.}$$

iii. As motion is SHM,  $v = 0$  when  $x$  is a min/max. Thus

$$0 = 32 + 8x - 4x^2$$

$$= -4(x^2 - 2x - 8)$$

$$= -4(x-4)(x+2)$$

$$\therefore -2 \leq x \leq 4 \quad \text{❶}$$

iv. The particle starts at  $x = 3$ , moving to the left and reaching a maximum velocity of  $v = 6$  when  $x = 1$ , then decelerating to a stop at  $x = -2$  ❶

b)

$$\begin{aligned} \text{i. } \frac{d(x \sin^{-1} x)}{dx} &= \sin^{-1} x + \frac{x}{\sqrt{1-x^2}} \quad \text{❶} \\ \sin^{-1} x &= \frac{d(x \sin^{-1} x)}{dx} - \frac{x}{\sqrt{1-x^2}} \end{aligned}$$

Integrating both sides:

$$\begin{aligned} \int \sin^{-1} x \, dx &= \int \frac{d(x \sin^{-1} x)}{dx} dx - \int \frac{x}{\sqrt{1-x^2}} dx \\ &= x \sin^{-1} x - \int \frac{x}{\sqrt{1-x^2}} dx \quad \text{❶} \end{aligned}$$

**Marking****Comments**

Solution/description must include the effects crossing  $x = 1$ .

i. Generally answered well. Students need to explicitly state it's SHM because it's in the form  $\ddot{x} = -n^2x$ . A mark was deducted if both  $n$  and the C.O.M were not stated explicitly

ii. Generally answered well

iii. Answered poorly. Many students stated only values of  $x$  at the endpoints instead of a range of values

iv. Answered poorly. Position of particle at key points need to be stated along with velocity and acceleration. Most gave insufficient answers

i. Not answered very well. Students need to state the eqn. for evaluation. Don't include sign of integration in front of  $x \sin^{-1} x$ .

## Marking

## Comments

$$\begin{aligned}
 \text{ii. } \therefore \int_0^1 \sin^{-1} x \, dx &= \left[ x \sin^{-1} x \right]_0^1 - \int_0^1 \frac{x}{\sqrt{1-x^2}} dx \\
 &= \left[ x \sin^{-1} x \right]_0^1 + \frac{1}{2} \int_0^1 \frac{-2x}{\sqrt{1-x^2}} dx \\
 &= 1 \cdot \sin^{-1}(1) - 0 \cdot \sin^{-1}(0) + \frac{1}{2} \cdot \frac{2}{1} \left[ \sqrt{1-x^2} \right]_0^1 \text{ ①} \\
 &= \frac{\pi}{2} - 0 + (0-1) \\
 &= \frac{\pi}{2} - 1 \text{ ①}
 \end{aligned}$$

c)

$$\text{i. } x = Vt \cos \beta \text{ ①}$$

$$y = h + Vt \sin \beta - \frac{1}{2}gt^2 \text{ ①}$$

ii. At time T:

For object projected from origin O:

$$x_O = UT \cos \alpha \quad y_O = UT \sin \alpha - \frac{1}{2}gT^2$$

For object projected from A:

$$x_A = VT \cos \beta \quad y_A = h + VT \sin \beta - \frac{1}{2}gT^2$$

Also at this time T:  $x_O = x_A$  and  $y_O = y_A$ , thus (noting no V in the final required solution)

$$UT \cos \alpha = VT \cos \beta$$

$$\begin{aligned}
 V &= \frac{UT \cos \alpha}{T \cos \beta} \\
 &= \frac{U \cos \alpha}{\cos \beta} \quad \{1\} \text{ ①}
 \end{aligned}$$

$$UT \sin \alpha - \frac{1}{2}gT^2 = h + VT \sin \beta - \frac{1}{2}gT^2$$

$$UT \sin \alpha = h + VT \sin \beta$$

$$h = UT \sin \alpha - VT \sin \beta \quad \{2\} \text{ ①}$$

{1} into {2}:

ii. Many careless errors.  
Some students were unable to correctly integrate the term  $\frac{x}{\sqrt{1-x^2}} dx$

i. Answered well

ii. Finding the Cartesian equation by eliminating  $T$  was not necessary since it appears in the eqn to be proven.  $h$  was added to the horizontal ( $x$ ) component instead of the vertical.  
If students didn't explicitly show the expanded form of  $\sin(\alpha - \beta)$  a mark was deducted

$$\begin{aligned}h &= UT \sin \alpha - \left( \frac{U \cos \alpha}{\cos \beta} \right) T \sin \beta \textbf{①} \\&= T \left( U \sin \alpha - \frac{U \cos \alpha \sin \beta}{\cos \beta} \right) \\&= TU \left( \frac{\sin \alpha \cos \beta - \sin \beta \cos \alpha}{\cos \beta} \right) \\h \cos \beta &= TU (\sin(\alpha - \beta)) \textbf{①} \\T &= \frac{h \cos \beta}{U (\sin(\alpha - \beta))} \quad \text{as reqd.}\end{aligned}$$

Marking

Comments

**Question 11** (d) – using the transformation:

$$1 + \cos 2x = \sin 2x$$

$$1 = \sin 2x - \cos 2x$$

$$(i) R \sin 2x \cos \theta - R \sin \theta \cos 2x = R \sin(2x - \theta)$$

$$R^2 = 1^2 + 1^2 \text{ and } \tan \theta = \left(\frac{1}{1}\right)$$

$$R = \sqrt{2}$$

$$\theta = \tan^{-1}(1)$$

$$= \frac{\pi}{4} \text{ ①}$$

$$\sqrt{2} \sin\left(2x - \frac{\pi}{4}\right) = 1$$

$$\sin\left(2x - \frac{\pi}{4}\right) = \frac{1}{\sqrt{2}}$$

$$2x - \frac{\pi}{4} = n\pi + (-1)^n \sin^{-1}\left(\frac{1}{\sqrt{2}}\right) \text{ ①}$$

$$2x = n\pi + \frac{\pi}{4} + (-1)^n \frac{\pi}{4}$$

$$x = n\frac{\pi}{2} + \frac{\pi}{8} + (-1)^n \frac{\pi}{8}$$

$$= (4n+1 + (-1)^n) \frac{\pi}{8} \text{ ②}$$

with  $n = 0, \pm 1, \pm 2, \dots$  for the general solution. With  $n$  even:  $n = 2m$  gives:

$$x = (8m+1+1) \frac{\pi}{8}$$

$$= (4m+1) \frac{\pi}{4}$$

$$= \dots, \frac{-3\pi}{4}, \frac{\pi}{4}, \frac{5\pi}{4}, \frac{9\pi}{4}, \dots$$

$$= \frac{\pi}{4}, \frac{5\pi}{4} \text{ (for } 0 \leq x \leq 2\pi) \text{ ①}$$

For  $n$  odd,  $n = 2m+1$  gives:

$$x = (4(2m+1)+1-1) \frac{\pi}{8}$$

$$= (2m+1) \frac{\pi}{2}$$

$$= \dots, \frac{-3\pi}{2}, \frac{\pi}{2}, \frac{3\pi}{2}, \frac{5\pi}{2}, \dots$$

$$= \frac{\pi}{2}, \frac{3\pi}{2} \text{ (for } 0 \leq x \leq 2\pi) \text{ ①}$$

$$1 + \cos 2x = \sin 2x$$

$$\cos 2x - \sin 2x = -1$$

$$(ii) R \cos 2x \cos \theta + R \sin 2x \sin \theta = R \cos(2x + \theta)$$

$$R^2 = 1^2 + 1^2 \text{ and } \tan \theta = \left(\frac{1}{1}\right)$$

$$R = \sqrt{2}$$

$$\theta = \tan^{-1}(1)$$

$$= \frac{\pi}{4} \text{ ①}$$

$$\sqrt{2} \cos\left(2x + \frac{\pi}{4}\right) = -1$$

$$\cos\left(2x + \frac{\pi}{4}\right) = \frac{-1}{\sqrt{2}}$$

$$2x + \frac{\pi}{4} = 2n\pi \pm \cos^{-1}\left(\frac{-1}{\sqrt{2}}\right) \text{ ①}$$

$$2x = 2n\pi - \frac{\pi}{4} \pm \frac{3\pi}{4}$$

$$x = n\pi - \frac{\pi}{8} \pm \frac{3\pi}{8}$$

$$= (8n-1 \pm 3) \frac{\pi}{8} \text{ ②}$$

with  $n = 0, \pm 1, \pm 2, \dots$  for the general solution.

$$x = \dots, \frac{-4\pi}{8}, \frac{2\pi}{8}, \frac{4\pi}{8}, \frac{10\pi}{8}, \frac{12\pi}{8}, \frac{18\pi}{8}, \dots$$

$$= \frac{\pi}{4}, \frac{\pi}{2}, \frac{5\pi}{4}, \frac{3\pi}{2} \text{ (for } 0 \leq x \leq 2\pi) \text{ ① ②}$$

**Please note:**

- how much easier a transformation into  $\cos$  is compared to  $\sin$
- ① and ② are sometimes referred to as **generators** for the fractional value of  $\pi$  in the general solutions (again, note the easier version that  $\cos$  has over  $\sin$ ). Some students attempted to use the generators, only a couple were successful – you must pay attention to the required range. Successful students usually used something such as:

$$0 \leq x \leq 2\pi$$

$$0 \leq 2x \leq 4\pi$$

$$\frac{-\pi}{4} \leq 2x - \frac{\pi}{4} \leq 4\pi - \frac{\pi}{4}$$

to ensure their answers fell into the required range.